

```

# Install dataset (only first time in Colab)
!pip install nltk

# Import libraries
import nltk
import random
import pandas as pd

from nltk.corpus import movie_reviews
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
from sklearn.metrics import ConfusionMatrixDisplay
import matplotlib.pyplot as plt

ConfusionMatrixDisplay.from_predictions(y_test, y_pred)
plt.show()

# Download dataset
nltk.download('movie_reviews')

# Load data
documents = []
for category in movie_reviews.categories():
    for fileid in movie_reviews.fileids(category):
        documents.append((" ".join(movie_reviews.words(fileid)), category))

# Convert to DataFrame
df = pd.DataFrame(documents, columns=['review', 'sentiment'])

# Convert labels (pos=1, neg=0)
df['sentiment'] = df['sentiment'].map({'pos':1, 'neg':0})

# Split data (FIXED for stable accuracy)
X_train, X_test, y_train, y_test = train_test_split(
    df['review'], df['sentiment'],
    test_size=0.2,
    random_state=42,
    stratify=df['sentiment']
)

# TF-IDF
vectorizer = TfidfVectorizer(stop_words='english', max_features=5000)
X_train_tfidf = vectorizer.fit_transform(X_train)
X_test_tfidf = vectorizer.transform(X_test)

# Model
model = LogisticRegression(max_iter=300)
model.fit(X_train_tfidf, y_train)

# Prediction
y_pred = model.predict(X_test_tfidf)

# Accuracy
print("Accuracy:", accuracy_score(y_test, y_pred))

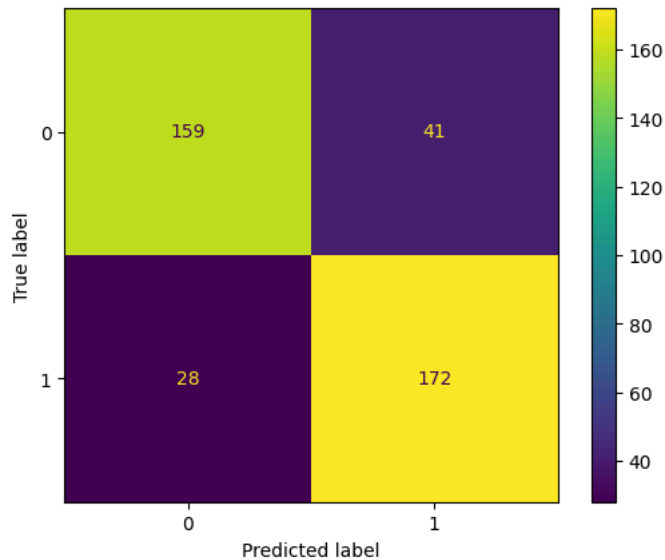
# Report
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

# Custom test
sample = ["This movie was absolutely fantastic and amazing"]
sample_tfidf = vectorizer.transform(sample)

prediction = model.predict(sample_tfidf)
print("\nCustom Review Prediction:", "Positive" if prediction[0]==1 else "Negative")

```

```
Requirement already satisfied: nltk in /usr/local/lib/python3.12/dist-packages (3.9.1)
Requirement already satisfied: click in /usr/local/lib/python3.12/dist-packages (from nltk) (8.3.3)
Requirement already satisfied: joblib in /usr/local/lib/python3.12/dist-packages (from nltk) (1.5.3)
Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.12/dist-packages (from nltk) (2025.11.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.12/dist-packages (from nltk) (4.67.3)
```



```
[nltk_data] Downloading package movie_reviews to /root/nltk_data...
[nltk_data] Package movie_reviews is already up-to-date!
Accuracy: 0.8275
```

Classification Report:

	precision	recall	f1-score	support
0	0.85	0.80	0.82	200
1	0.81	0.86	0.83	200
accuracy			0.83	400
macro avg	0.83	0.83	0.83	400
weighted avg	0.83	0.83	0.83	400

Custom Review Prediction: Positive

✦ Title

Sentiment Analysis using NLP

✦ Objective

To classify customer reviews as Positive or Negative using NLP techniques.

✦ Libraries Used

nltk, pandas, scikit-learn

✦ Dataset

Movie Reviews dataset from NLTK

Result:

Sentiment analysis helps in understanding customer opinions.

Using TF-IDF and Logistic Regression gives good performance.

Double-click (or enter) to edit

Conclusion:

Sentiment analysis helps in understanding customer opinions.

Using TF-IDF and Logistic Regression gives good performance.

